

EXECUTIVE SUMMARY

The AI Pharma Challenge

The Data Readiness Strategy for AI at Scale

AI Data Readiness for
Global Pharmaceutical Enterprises

March 2026 · Executive Edition

The Challenge

Pharmaceutical has more to gain from AI than almost any other industry. The question facing global top pharma today is not whether to invest, but how to deploy AI at full enterprise scale—faster and more cost-effectively than the competition—without falling into the traps that turn ambitious programs into siloed, overpriced, underpowered installations.

The traps are real: 42% of enterprise AI initiatives failed to meet ROI expectations in 2025. 95% of manufacturing AI pilots fail to deliver measurable business value. 68% of pharma tech executives cite data quality and governance as the primary reason AI fails. Independent CIO research confirms the pattern: 93% of organizations are actively deploying AI, but only 21% report success rates above 80%. The gap is implementation and infrastructure, not technology.

\$610B

2026 hyperscaler CapEx

42%

AI initiatives miss ROI

95%

Mfg pilots fail at scale

68%

Cite data as root cause

The External Pressure

Cloud and neocloud providers are engaged in a historical race to capacity, committing over \$610 billion in 2026 CapEx—acquiring land, energy contracts, and grid connections at a pace that has turned them into extraction businesses. GPU capacity is physically scarce, pricing is structurally inflationary, and the most cloud-dependent organizations are the most exposed to rationing, repricing, and lock-in. For pharmaceutical enterprises with multi-year AI programs and sovereignty requirements across multiple jurisdictions, this creates an urgent strategic question: where should AI infrastructure run, who should own it, and on what economic terms?

The Architecture Decision

Across five geographies assessed in the full report (USA, Switzerland, Germany, France, China), the same architectural question recurs: what belongs in the cloud, what belongs in colocation, and what belongs on-premises? The answer is not ideological—it is economic, operational, and regulatory:

- Cloud — for elastic, variable workloads: burst prototyping, subsidiary analytics, geographically distributed teams. Not for the predictable base.
- Colocation — for AI-class power density (40–80 kW per rack) that campus facilities cannot support. Purpose-built cooling, redundant power, cost predictability.
- On-premises — for the predictable core: proprietary drug discovery pipelines, clinical data under sovereignty constraints, manufacturing AI tightly coupled to production lines.

The critical insight: a data platform that eliminates refresh cycles, runs subscription-based, and operates identically across all three tiers neutralizes the primary objection to on-premises infrastructure—technology obsolescence—while retaining sovereignty, efficiency, and cost control.

The Data Layer: Where AI ROI Is Determined

AI pilot failure in pharma traces overwhelmingly to the data layer. GPU utilization rates of 20–40% are endemic where storage cannot sustain the mixed-size I/O patterns that define AI workloads. Genomics files range from kilobyte metadata to multi-gigabyte BAM/CRAM alignments. Digital pathology generates 1–10 GB per study. Model training requires frequent checkpoint writes during multi-day runs. The data platform must handle all of these simultaneously, without tuning or tiering.

What the Platform Must Deliver

- Performance at AI scale: Sustained throughput for training, sub-millisecond latency for inference, parallel I/O across the full array.
- Capacity without compromise: Petabyte-scale from a single namespace. Non-disruptive expansion. Cost-optimized tiers for raw data, active training, and long-term archival.
- Guaranteed availability: Six-nines uptime. No planned downtime for upgrades. GxP-validated, non-disruptive operations.
- Cyber resilience: Immutable snapshots, rapid recovery SLAs, ransomware-proof architecture for protecting pharmaceutical IP worth billions.
- Sovereignty and compliance: Data never leaves the jurisdiction. On-premises encryption. Full audit trails for EU GMP Annex 11, HIPAA, PIPL.
- Economic predictability: Subscription-based, no forklift upgrades, no end-of-life write-downs. Storage as perpetually current OpEx.
- Operational simplicity: Single management plane across geographies. API-first. Integrated with ServiceNow and existing ITSM workflows.

Everpure: The Platform

Everpure is the only data platform that satisfies all seven requirements simultaneously. Its Enterprise Data Cloud (EDC) architecture consolidates block, file, and object storage across on-premises, hybrid, and cloud environments under a single intelligent control plane. The portfolio spans extreme performance (FlashBlade//S, FlashArray//XL) through cost-optimized capacity (FlashBlade//E, FlashArray//E), unified under Purity OS and managed through Everpure Fusion.

Key differentiators for pharmaceutical AI:

- NVIDIA DGX SuperPOD certified — AIRI reference architecture eliminates DIY integration risk.
- GPUDirect Storage — bypasses CPU bottleneck, lifting GPU utilization from 20–40% to near 100%.
- Evergreen//One — consumption model with fixed \$/GiB, buffer capacity included, no overprovisioning penalties. Converts CapEx to OpEx. Outperforms cloud NPV by 2–3x over five years.
- SafeMode — immutable snapshots that survive compromised admin credentials. Rapid-recovery SLAs.
- Everpure Fusion — as-code management across all geographies from a single pane. 50–70% reduction in admin overhead.
- Portworx — Kubernetes data services extending platform consistency to containerized AI workloads, eliminating cloud-native to storage friction for data science teams.
- 1touch (Kontxtual™) — contextual data intelligence: automated discovery, classification, and semantic enrichment across all environments (cloud, on-prem, SaaS, mainframe). 98.6% accuracy validated by Tolly Group. Powers IBM Guardium Discover and Classify globally. Transforms raw data into AI-ready assets at the source; addresses GDPR/PIPL/GxP sovereignty with continuous compliance mapping.

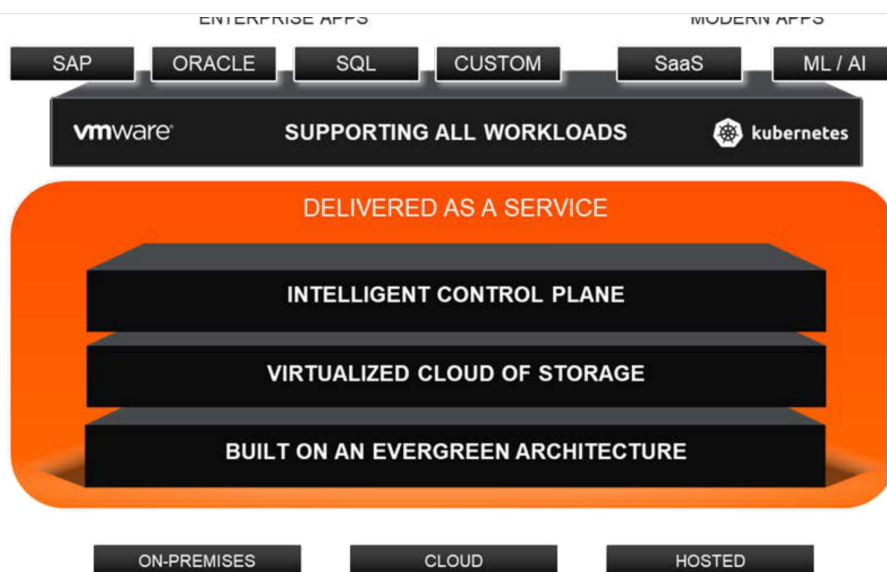


FIGURE 2 The Enterprise Data Cloud architecture.

The Enterprise Data Cloud Architecture. Source: Everpure EDC Architecture White Paper.

From Awareness to Action

- 30 Days — Establish the Facts: Instrument GPU utilization across every AI cluster. Profile storage I/O. Conduct cloud account census. Map data sovereignty exposure. Deploy automated data discovery across a representative subset of environments—catalogue what sensitive data exists, where it resides, and how it is classified. These exercises cost nothing except executive attention and produce the factual base for every subsequent decision.
- 60–120 Days — Build the Business Case: Model TCO: current infrastructure plus GPU idle time versus a purpose-built AI data platform at full utilization. Define the workload allocation framework: cloud (elastic), colocation (power-constrained), on-premises (predictable core). Evaluate data platforms against the nine criteria.
- 90–150 Days — Design and Pilot: Select 2–3 representative AI workloads spanning drug discovery, clinical inference, and manufacturing. Deploy Everpure in a controlled pilot measuring GPU utilization, I/O throughput, and operational simplicity. Validate Evergreen//One economics against cloud benchmarks.
- 150–240 Days — Deploy at Scale: Extend to production across prioritized geographies. Activate SafeMode, configure Everpure Fusion for cross-geography governance, establish Portworx for containerized workloads. Decommission legacy storage. Reallocate cloud spend.

Conclusion

The pharmaceutical industry's AI infrastructure challenge is not a technology problem—it is a data readiness and strategic allocation problem. The hyperscaler capacity race has created structural inflation in cloud GPU pricing. Regulatory fragmentation across geographies demands sovereign infrastructure. And the 42% AI ROI failure rate traces directly to data foundations that were never designed for the sustained, mixed-workload, petabyte-scale demands of enterprise AI.

The organizations that will lead in pharmaceutical AI are those that make four decisions now: own the predictable core, rent the elastic margin, build on a data platform that never becomes obsolete, and treat infrastructure economics as a competitive weapon—not a cost center. Everpure's Enterprise Data Cloud is the platform built for exactly this moment.



Everpure — 100% NVMe, Simplicity, Efficiency, 99.9999% Availability, AI/ML Management, Security, Encryption, Data Reduction

This executive summary is drawn from the full 24-page report, which includes detailed hyperscaler CapEx analysis, five-country regulatory assessment, cloud reservation structure analysis, storage requirements framework, implementation timeline, and self-assessment scorecard. Contact your Everpure account team for the complete document.